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Artificial intelligence in fetal brain imaging: Advancements, challenges, and multimodal approaches for biometric and structural analysis

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ABSTRACT

Artificial intelligence (AI) is transforming fetal brain imaging by addressing key challenges in diagnostic accuracy, efficiency, and data integration in prenatal care. This review explores AI's application in enhancing fetal brain imaging through ultrasound (US) and magnetic resonance imaging (MRI), with a particular focus on multimodal integration to leverage their complementary strengths. By critically analyzing state-of-the-art AI methodologies, including deep learning frameworks and attention-based architectures, this study highlights significant advancements alongside persistent challenges. Notable barriers include the scarcity of diverse and high-quality datasets, computational inefficiencies, and ethical concerns surrounding data privacy and security. Special attention is given to multimodal approaches that integrate US and MRI, combining the accessibility and real-time imaging of US with the superior soft tissue contrast of MRI to improve diagnostic precision. Furthermore, this review emphasizes the transformative potential of AI in fostering clinical adoption through innovations such as real-time diagnostic tools and human-AI collaboration frameworks. By providing a comprehensive roadmap for future research and implementation, this study underscores AI's potential to redefine fetal imaging practices, enhance diagnostic accuracy, and ultimately improve perinatal care outcomes.

1. Introduction

Fetal health complications, particularly intrauterine growth restriction (IUGR), have significant implications for postnatal development and long-term outcomes. Affecting approximately 30 % of neonates, intrauterine growth restriction is a major contributor to perinatal morbidity and mortality, underscoring the critical need for early diagnosis and intervention [1]. Effective perinatal care relies on advanced imaging techniques to monitor fetal growth and development, with ultrasound (US) and magnetic resonance imaging (MRI) serving as the primary diagnostic modalities.

US imaging is currently the gold standard medical imaging tool for fetal health assessment due to its real-time imaging capability, accessibility, and established safety profile [2]. It enables continuous monitoring of fetal development, early detection of abnormalities, and detailed visualization of fetal anatomy. Recent advancements, such as three-dimensional (3D) US, have enhanced diagnostic capabilities by providing more comprehensive and detailed information compared to traditional two-dimensional (2D) techniques [3,4]. Despite challenges such as fetal movement, variations in maternal abdominal wall thickness, and observer-dependent variability, obstetric ultrasonography remains the most accessible, non-invasive, and widely used method for evaluating fetal growth and well-being. Conversely, MRI provides superior soft tissue contrast and higher spatial resolution, making it particularly valuable for detailed neuroanatomical assessments and complex cases. However, MRI is constrained by higher costs, limited accessibility, and susceptibility to motion artifacts [5]. The strengths and limitations of both modalities highlight the need for advanced computational solutions to improve diagnostic accuracy and workflow efficiency.

Artificial intelligence (AI), particularly deep learning (DL) approaches, has emerged as a transformative tool in fetal brain image analysis. AI applications in the US and MRI have demonstrated significant clinical utility in anatomical plane detection and brain structure segmentation [6–9]. Notably, fetal multiplanar neurosonography, encompassing axial, coronal, and sagittal views, combined with MRI,

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has provided incremental diagnostic value over standard axial US planes for detecting brain damage [10]. Despite these advancements, the widespread clinical adoption of AI tools remains constrained by challenges related to model reproducibility, generalizability across diverse populations, and the interpretability of results. While most AI models are tailored for single modalities, emerging research is exploring multimodal data integration to harness the complementary strengths of US and MRI (see Fig. 1). This integrative approach holds great promises for enhancing diagnostic precision, enabling personalized assessments, and advancing the understanding of fetal brain development.

This study examines state-of-the-art clinical applications of AI technologies in the US, MRI, and multimodal imaging, laying the foundation for future integrative systems. The study provides a comprehensive overview of AI applications in fetal imaging by evaluating how AI technologies enhance diagnostic capabilities and assessing their potential for cross-modality integration. It emphasizes current limitations while highlighting directions for future research. The study addresses the following key research questions.

1. What publicly accessible datasets are commonly utilized in the development of AI algorithms for fetal brain imaging using US and/or MRI?
2. Which AI algorithms are most frequently employed for the detection of standard planes in fetal US and/or MRI?
3. What are the predominant AI algorithms used for the segmentation of fetal brain structures from US and/or MRI data?
4. Which AI algorithms are typically applied for the estimation of biometric parameters in fetal US and/or MRI?
5. What strategies can enhance the effective integration of AI into the analysis of fetal brain imaging within clinical settings?

The rest of this paper is structured as follows: Section 2 outlines the research methodology. Section 3 introduces the fundamental AI algorithms and evaluation metrics applied in these studies. Section 4 critically evaluates AI-driven advancements in fetal US diagnostics. Section 5 focuses on AI applications in fetal MRI, addressing three key areas: brain segmentation, multi-tissue annotation pipelines, and automated biometric analysis. Section 6 explores innovative multimodal imaging approaches that combine US and/or MRI data. Section 7 discusses persistent challenges and presents evidence-based priorities for future research. Finally, Section 8 concludes the study.

2. Methods

This review focused on studies that applied AI techniques to analyze fetal imaging, particularly US, MRI, and multimodal data. The methodology followed a structured approach to ensure a comprehensive review of literature.

A comprehensive literature search was conducted across multiple databases, including PubMed, Scopus, Google Scholar, and IEEE Xplore, using carefully selected keywords. The search was restricted to studies published in English between January 2010 and March 30, 2025, to capture the most recent advancements in AI-based fetal imaging. The decision to focus on studies from 2010 onward was based on the notable surge in AI-driven research in fetal imaging during this period, ensuring that our review reflects contemporary developments in this rapidly evolving field.

The search strategy incorporated a combination of terms related to fetal imaging and AI techniques. Specifically, we used the following key terms: ((“fetal”) OR (“fetus”)) AND ((“ultrasound”) OR (“ultrasonography”) OR (“MRI”) OR (“magnetic resonance imaging”) OR (“multimodal”) OR (“multi-modal”)) AND ((“artificial intelligence”) OR (“AI”) OR (“deep learning”) OR (“machine learning”)). Boolean operators were employed to refine the search, with the OR operator used to combine terms within each category and the AND operator used to link terms across categories.

Studies were included if they incorporated multimodal data, such as a combination of US and MRI, along with clinical parameters like laboratory results or vital signs. The review considered peer-reviewed articles, dissertations, book chapters, and conference proceedings published between January 2010 and March 2025, provided they focused on AI applications in fetal brain imaging using US, MRI, or multimodal approaches. Exclusion criteria were applied to eliminate studies unrelated to fetal imaging or lacking AI-based analysis. Additionally, abstracts, non-English publications, and studies published before 2010 were excluded to maintain relevance and rigor. The study selection process was conducted independently by one author, with any disagreements resolved through discussion and consensus among all authors.

The initial database search retrieved 375 records (PubMed: 76; Scopus: 150; Google Scholar: 55; Web of Science: 94). After removing duplicates and screening titles and abstracts, 162 studies met the predefined criteria for full-text review. The selection process, including reasons for exclusion, is detailed in Figs. 2 and 3. Among the final selected studies, the literature distribution revealed distinct methodological focuses: 118 studies primarily utilized US imaging, 40 studies

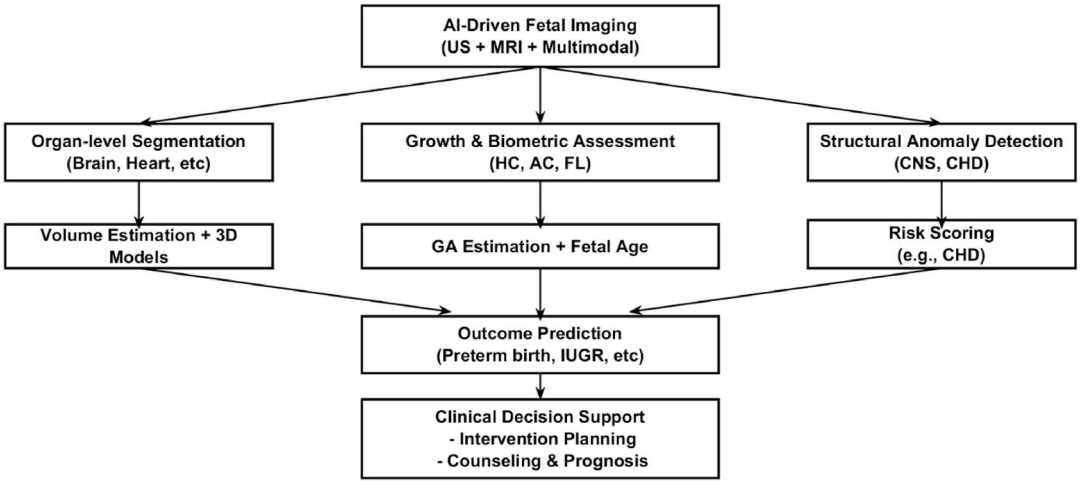


Fig. 1. Flow chart of clinical applications of AI-driven fetal imaging. AC: Abdominal Circumference, HC: Head Circumference, FL: Femur Length, CNS: CNS: Central Nervous System, CHD: Congenital Heart Disease, GA: Gestational Age, IUGR: Intrauterine Growth Restriction.

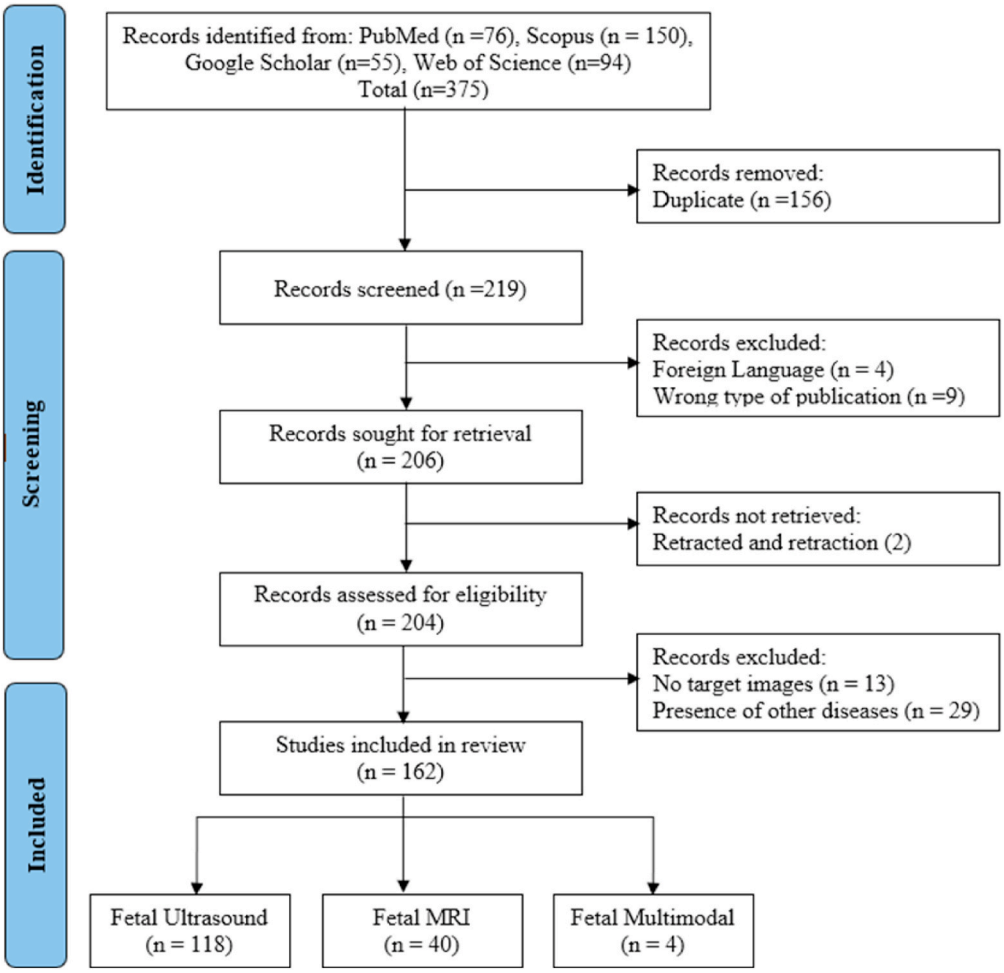


Fig. 2. PRISMA diagram showing literature study inclusion process.

employed MRI, and 4 studies incorporated multimodal imaging approaches.

3. AI algorithms

AI has revolutionized fetal brain imaging through advancements in segmentation, classification, and anomaly detection across US and MRI modalities. These AI-driven innovations have significantly improved diagnostic accuracy, minimized observer variability, and automated traditionally labor-intensive image analysis processes. The clinical implications are profound, as these technologies enhance workflow efficiency and deliver precise quantitative assessments that empower prenatal diagnoses and facilitate informed decision-making.

3.1. Machine learning

Machine learning (ML) algorithms have become fundamental tools in fetal brain imaging research, primarily implemented through three methodological approaches. Supervised learning techniques, including support vector machines and random forests, have proven particularly effective for segmentation tasks and anomaly detection [11]. These methods rely on large annotated datasets to establish relationships between imaging features and clinical outcomes, though their performance can be limited by requirements for extensive labeled data and susceptibility to overfitting.

Unsupervised learning methods such as k-means clustering and autoencoders offer valuable alternatives by identifying patterns in unlabeled data, making them particularly useful for distinguishing normal from abnormal neuroanatomy [12]. Recent innovations include the application of generative adversarial networks to create synthetic training data that enhances model robustness. Reinforcement learning has emerged as a promising approach for optimizing imaging protocols through adaptive, patient-specific adjustments, though its clinical implementation faces challenges related to computational demands and reward function design [13].

3.2. Deep learning

Deep learning (DL), as a specialized branch of ML, uses multiple layers of artificial neural networks (ANNs) to automatically learn from raw data, eliminating the need for manual feature extraction [14,15]. ANNs mimic human neurons and process data non-linearly, making DL highly effective [16]. Its rapid growth is fueled by advancements in computing power, improved learning algorithms, and access to large datasets [17]. These factors have enabled DL to achieve significant breakthroughs across diverse fields, establishing it as a leading AI approach for complex tasks.

Convolutional Neural Networks (CNNs) have emerged as one of the most powerful DL tools for medical image analysis [18–23]. A typical CNN architecture consists of convolutional layers (feature extraction), pooling layers (downsampling and overfitting control), fully connected layers (classification), and non-linearity layers (activation functions). These layers enable CNNs to learn hierarchical and abstract features from images, making them highly effective for tasks such as image classification and feature extraction. In fetal US imaging, CNNs have demonstrated significant potential in several critical tasks, including the

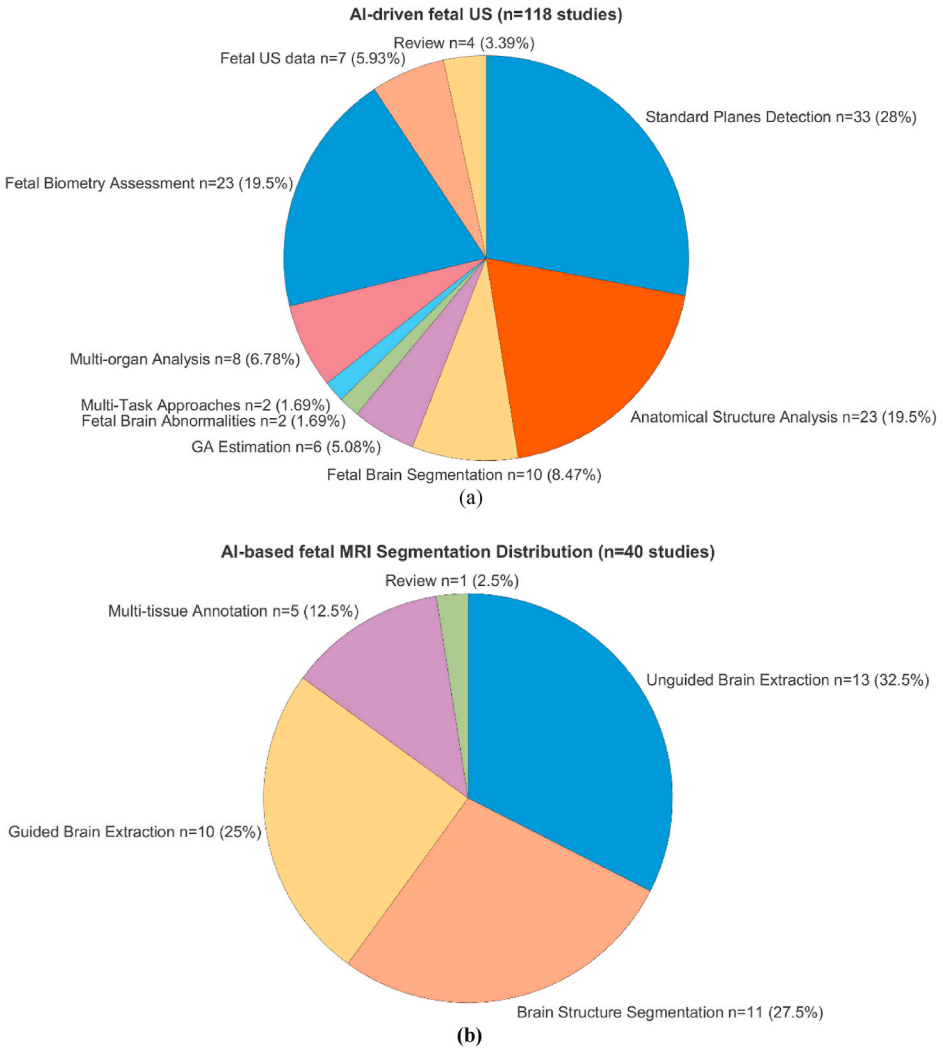


Fig. 3. Pie chart literature research selection (a) AI-driven fetal US, (b) AI-driven fetal MRI segmentation research.

recognition, detection, and localization of standard planes. For instance, CNNs can effectively recognize fetal standard planes such as the transventricular and transcerebellar planes, and perform biometric measurements with high accuracy. These capabilities are particularly valuable in prenatal diagnostics, where precise identification of standard planes is essential for assessing fetal development and detecting potential abnormalities. Although clinical applications of CNNs have been explored for fetal plane detection, the recognition of abnormalities remains a challenging task due to a lack of very large fetal anomaly datasets.

Transfer learning was effectively employed to adapt models pretrained on general medical imaging datasets for fetal brain imaging applications. This approach proves particularly beneficial in contexts where labeled fetal datasets are limited, but it encounters challenges related to domain adaptation and dataset bias [24,25].

3.3. Evaluation metrics

The rigorous evaluation of AI algorithms requires performance metrics that are appropriately tailored to specific tasks, such as classification, segmentation, localization, and regression [26]. For classification models, metrics like accuracy, sensitivity, specificity, precision, and F1 score are commonly employed to assess various facets of diagnostic performance. Accuracy gauges the overall correctness of predictions, while recall (or sensitivity) measures the model's ability to identify all relevant instances. Specificity evaluates the model's capacity to correctly recognize negative cases, and precision quantifies the proportion of true positives among all positive predictions. The F1 score, which harmonizes precision and recall, serves as a comprehensive metric to capture the balance between these two aspects of performance.

For segmentation and localization tasks, widely used metrics include the Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and Hausdorff Distance (HD), which measure spatial accuracy and overlap between predicted and true regions. Precision is often employed as a localization metric, while Mean Average Precision (mAP) is a standard evaluation metric for detection performance.

For regression tasks, metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE), are used to assess the accuracy of mask contours generated by segmentation algorithms. Additionally, the Area Under the Curve (AUC) quantifies the model's ability to distinguish between positive and negative instances. A higher AUC indicates better overall model performance. Together, these metrics facilitate quantitative evaluation of model performance, enabling informed model selection and optimization.

3.4. Datasets

Tables 1 and 2 highlight the publicly available fetal US and MRI datasets, which have become invaluable resources for advancing AI research in prenatal diagnostics [27–33]. These datasets play a critical

Table 1
Publicly available fetal US datasets used in the articles reviewed for this study.

Ref	Dataset	Task(s)	Training set (images)	Testing set (images)	GA (weeks)
[27]	Challenge US	Segmentation & measurement	270	~	21, 28, 33
[28]	HC18 challenge dataset	HC measurement	999	335	12~35
[29]	FHSPs	Standard plane classification	7129	5271	18~40
[30]	A-AFMA US	Amniotic fluid detection, maternal bladder, and landmarks identification	~	~	~
[31]	FPUS23	Diagnostic planes, fetus orientation, anatomical features, and bounding boxes	15728	~	~
[32]	PSFHS	Fetal head segmentation	4000	Set 1: 401, Set 2: 700	~
[33]	ACOUSLIC-AI	AC estimation	300 cases	290 cases	~

HC: Head circumference; AC: Abdominal circumference; GA: Gestational age.

Table 2
Publicly available fetal MRI datasets used in the articles reviewed for this study.

Ref	Template	Isotropic resolution (mm)	Region of interest	GA (weeks)	Original cohort
[34]	T2	0.5	4	21–24	20
[35]	T2	0.5	7	20–34	50
[36]	T2	0.5	7	20–35	80
[37]	T2	0.8	124	21–38	81
[38]	T2	0.8	124	22–35	89
[39]	T2	0.8	8	21–34	37
[40]	T2	0.8	85	23–38	90
[41]	T2	0.86	9	21–38	81
[41]	T2	0.86	9	33–38	20
[42]	T2	0.86	4	23–37	80
[43]	T1, T2	0.5	19	21–36	187
[43]	diffusion weighted imaging, fractional anisotropy, mean diffusivity, radial diffusivity, fibre orientation distribution	0.5	19	21–36	187
[44]	fractional anisotropy, corrected fractional anisotropy, mean diffusivity	0.75	–	22–38	67
[45]	tensor, fractional anisotropy, corrected fractional anisotropy, mean diffusivity, fibre orientation distribution, radial diffusivity, axial diffusivity	1.2	–	24–38	89

GA: Gestational age.

role in the development and validation of automated analysis tools, driving progress in fetal brain imaging and contributing to improved diagnostic accuracy and patient outcomes.

Among these resources, the HC18 dataset stands out as a pivotal contribution to fetal US research. It includes 999 training images and 335 testing images, specifically annotated for head circumference (HC) biometric analysis [27]. This dataset has proven instrumental in the development and validation of automated measurement algorithms for prenatal assessment.

In 2020, Burgos et al. [29] introduced a groundbreaking dataset of fetal US planes, which has since become a benchmark in prenatal imaging research. This curated dataset comprises 7129 training images and 5271 testing images, meticulously annotated and categorized into six anatomical groups: abdomen, brain, femur, thorax, maternal cervix, and other unspecified structures. Its systematic organization has enabled standardized evaluation of image analysis algorithms across key fetal anatomical regions.

4. Clinical applications of AI-driven fetal US

Fetal US is the current gold standard for prenatal anatomical evaluation, offering an ideal combination of safety, real-time imaging capabilities, and diagnostic reliability. Recent advancements in AI-driven fetal US have significantly improved capabilities across multiple domains. These include the automated identification of fetal standard planes, anatomical structure analysis, and automatic measurement of biometric parameters. As summarized in Tables 3–6, these breakthroughs have been demonstrated using both clinical and publicly available US datasets [27–33].

4.1. Identification of fetal standard planes

The International Society of Ultrasound in Obstetrics and Gynecology (ISUOG) guidelines emphasize the importance of acquiring essential standard planes for fetal evaluation, including the fetal brain (FBSP), abdomen (FASP), face (FFSP), femur (FFESP), and cardiac views, such as the 4-chamber view (4CH). Consistent acquisition of these planes is crucial as they provide the anatomical foundation for accurate biometric measurements and comprehensive assessment of fetal development and well-being. For example, FBSP evaluation involves the systematic analysis of three standardized axial planes: trans-ventricular (TV), transthalamic (TT), and trans-cerebellar (TC). These planes, along with their anatomical landmarks, enable clinicians to assess fetal brain morphology, detect structural anomalies, and obtain critical biometric data during routine ultrasound examinations.

Despite their clinical significance, precise identification of fetal standard planes presents challenges due to technical constraints, such as artifacts (e.g., reverberation and speckle noise), which degrade image resolution and obscure key anatomical features. Structural similarities between adjacent brain regions, particularly in the TV, TT, and TC planes, further complicate differentiation due to overlapping echogenic patterns from shared features like the skull and midline structures. Accurate acquisition requires advanced technical expertise with ultrasound equipment and a profound understanding of fetal neuroanatomy [26]. These challenges underscore the need for standardized protocols and decision-support tools to assist sonographers, especially in training environments or resource-limited settings.

Table 3 summarizes recently developed AI-based models for standard planes assessment, including localization [46–48], recognition [49–53], classification [29,54–62], detection [63–69], and multi-task solution [49,70–76]. Localization models have demonstrated significant improvements, achieving accuracies from 84.03 % (ResNet) to 95.2 % (coarse-to-fine multi-task learning) [46]. Ni et al. [46] pioneered the radial component model (RCM), which achieved an accuracy of 85.6 %. Chen et al. [47] enhanced performance to 90.4 % using domain-transferred deep convolutional neural networks (Deep CNN). Recognition models have also seen remarkable progress, driven by specialized architectural innovations. Models integrating large datasets and advanced architecture have achieved exceptional results, including CNN-19-GAP-TR's 96.32 % accuracy with 4849 training cases [51] and

Table 3
AI-driven US for standard plane assessment.

Ref	Model	Task	Plane	Training set (size)	Testing Set (size)	Performance metrics
[46]	Radial Component Model (RCM)	Localization	FASP	1995	223 videos	Acc = 85.6 %
[46]	Resnet	Localization	Multiple	10,433	4521	Acc = 84.03 %
[46]	Mobilenet_v3	Localization	Multiple	10,433	4521	Acc = 91.48 %
[46]	Efficientnet	Localization	Multiple	10,433	4521	Acc = 92.81 %
[46]	Coarse-to-fine multi-task learning	Localization	Multiple	10,433	4521	Acc = 95.2 %
[47]	Domain transferred deep CNN	Localization	FASP	11,942	8718	Acc: 90.4 %, 82.2 % for T-CNN, R-CNN
[48]	Multi-agent reinforcement learning (MARL)	Localization	Multiple	539 3D	144 3D	Uterus: Average Acc = 7.03o/1.59 mm, Fetal brain: Average Acc = 9.75o/1.19 mm
[49]	Cluster	Recognition	FBSP	18,000	6000	Acc = 60.7 %, Prec = 71.1 %, Recall = 74.1 %, F1 = 70.4 %
[49]	Cluster	Recognition	FBSP	720	240	Acc = 53.2 %, Prec = 71.5 %, Recall = 69.1 %, F1 = 70.4 %
[49]	Support vector machine (SVM)	Recognition	FBSP	18,000	6000	Acc = 78 %, Prec = 81.2 %, Recall = 84.6 %, F1 = 89.7 %
[49]	SVM	Recognition	FBSP	720	240	Acc = 88.2 %, Prec = 84.4 %, Recall = 95.6 %, F1 = 89.7 %
[49]	RCM	Recognition	FBSP	18,000	6000	Acc = 73.3 %, Prec = 84.5 %, Recall = 91.3 %, F1 = 86.3 %
[49]	RCM	Recognition	FBSP	720	240	Acc = 73.4 %, Prec = 78.5 %, Recall = 81.3 %, F1 = 85.1 %
[49]	Deep convolutional neural network (DCNN)	Recognition	FBSP	18,000	6000	Acc = 91 %, Prec = 85.5 %, Recall = 90.1 %, F1 = 90 %
[49]	DCNN	Recognition	FBSP	720	240	Acc = 83.3 %, Prec = 84.2 %, Recall = 88.7 %, F1 = 83.1 %
[49]	Transfer learning	Recognition	FBSP	720	240	Acc = 89.1 %, Prec = 85.3 %, Recall = 86.4 %, F1 = 90.1 %
[50]	Fisher Vector with RootSIFT and SVM	Recognition	FFSP	1735	~	Acc = 93.27 %, mAP = 99.19 %
[51]	CNN-19-GAP-TR	Recognition	FFSP	4849	2418	Acc = 96.32 %, Prec = 91.91 %, Recall = 96.92 %, F1 = 94.35 %
[52]	Multi-scale dense networks (MSDNet)	Recognition	FASP, FBSP, FFSP, 4CH	17,036	5678	Prec = 98 %
[53]	Standard plane recognition network (SPRNet)	Recognition	FASP, FBSP, FFASP, FFSP, 4CH	17,840	4455	Acc = 99 %
[54]	Differential-CNN	Classification	FBSP	24,000	6000	Acc = 93.11 %, Prec = 92.62 %, Recall = 92.39 %, F1 = 93.53 %
[55]	FetalBrainAwareNet	Classification	FBSP	2043	1406	Prec = 92.62 %, 85.51 %, 65.91 % for TC, TT, TV, Mean average Acc = 82.43 %
[56]	Contrastive Learning	Classification	FBSP	1543	1406	For 90 % of labeled patients, F1 = = 87.61 %, 85.53 %, 76.10 % for TC, TT, TV
[57]	GANs with Class Activation Mappings	Classification	FBSP	2043	1406	Prec = 86 %, 86 %, 66 % for TC, TT, TV, Recall = 86 %, 76 %, 84 % for TC, TT, TV, F1 = 86 %, 81 %, 74 % for TC, TT, TV
[58]	Generative adversarial networks (GANs)	Classification	FBSP	4276	4471	Acc = 81.5 %, AUC = 86.7 %, F1 = 80 %
[59]	Self-supervised learning	Classification	Multiple	22,757	5737	Acc = 70 %
[60]	Ensemble CNN (ECNN)	Classification	FASP, FBSP, FFESP, thorax, maternal cervix, other	9193	3064	Acc = 95.69 %, Prec = 94.02 %
[60]	GoogLeNet					Acc = 92.98 %, Prec = 91.03 %
[60]	MobileNet-v2					Acc = 93.08 %, Prec = 91.16 %
[60]	ResNet-101					Acc = 93.17 %, Prec = 91.31 %
[60]	SqueezeNet					Acc = 93.31 %, Prec = 91.91 %
[60]	ResNet-18					Acc = 93.44 %, Prec = 91.28 %
[60]	ShuffleNet					Acc = 93.47 %, Prec = 91.75 %
[60]	EfficientNet-b0					Acc = 93.73 %, Prec = 92.17 %
[60]	ResNet-50					Acc = 93.83 %, Prec = 91.86 %
[60]	DarkNet-53					Acc = 93.83 %, Prec = 91.84 %
[60]	AlexNet					Acc = 94.12 %, Prec = 92.42 %
[60]	DarkNet-19					Acc = 94.22 %, Prec = 92.46 %
[61]	Unsupervised domain adaptation with metric feature alignment for volumes (UDA-MetFA-V)	Classification	FASP, LVOT, RVOT, FFESP, Lips, 4CH	12,000	5500	Prec = 74.41 %
[29]	VGG-M	Classification	FASP, FBSP, FFESP, thorax, maternal cervix, other	7129	5271	Acc = 84.8 %
[29]	VGG-16					Acc = 92.1 %
[29]	MobileNet					Acc = 87.5 %
[29]	Inception-v3					Acc = 93.5 %

(*continued on next page*)

Table 3 (*continued*)

Ref	Model	Task	Plane	Training set (size)	Testing Set (size)	Performance metrics
[29]	ResNet-18					Acc = 92.5 %
[29]	ResNet-34					Acc = 92.5 %
[29]	ResNet-50					Acc = 93.1 %
[29]	ResNet-101					Acc = 93.4 %
[29]	ResNet-152					Acc = 92.8 %
[29]	SENet					Acc = 92.9 %
[29]	SE-ResNet-50					Acc = 93.3 %
[29]	SE-ResNet-101					Acc = 93.3 %
[29]	SE-ResNet-152					Acc = 92.7 %
[29]	DenseNet-121					Acc = 92.9 %
[29]	DenseNet-169					Acc = 93.6 %
[29]	Baseline1 (PCA + Boosting)					Acc = 54.7 %
[29]	Baseline2 (Hog + Boosting)					Acc = 68.6 %
[62]	3- residual blocks CNN	Classification	Fetal dataset includes abdomen, brain, femur, thorax, maternal cervix, and not a brain	26,034	26,032	Acc = 88.6 %
[62]	4-residual blocks CNN	Classification	FPSU23 dataset includes abdominal circumference, biparietal diameter, femur length, and no plan	2633	2632	Acc = 98.5 %
[63]	Temporal SonoEyeNet	Detection	FASP, FBSP, FFESP	280 videos	~	F1: 83.7 %, 89.9 %, 81.1 % for FASP, FBSP, FFESP
[64]	Multi-task SonoEyeNet	Detection	FASP	1292	324	Prec = 96 %, Rec = 96 %, F1 = 96 %
[65]	Faster R-CNN	Detection	FBSP	4800	1153	Average Acc = 80.18 %
[66]	CNN	Detection	FBSP	19,838	~	Acc = 95.3 %, DSC = 96.9 %
[67]	CNN, DenseNet-201, Inception-V4, ResNet-152, ResNet-18, ResNet-50, Xception	Detection	4CH	4260	1496	Acc = 95 %
[68]	Iterative transformation network (ITN)	Detection	FVSP, FCSP	50 3D	22 3D	TV: error of 3.83 mm/12.7^0, TC: error of 3.80 mm/12.6^0
[69]	CNN, RNN, LST, multi-task learning	Detection	FASP, FFASP, FFVSP	FASP (11,942), FFASP (13,091), FFVSP (12,343)	FASP (8,718), FFASP (2,278), FFVSP (2,252)	Acc: 94.1 %, 71.7 %, 84.6 % for FASP, FFASP, FFVSP
[70]	Multi-task faster R-CNN (MF R-CNN)	Detection and classification	FBSP	1451	320	mAP = 93 %
[71]	Xception	Detection and classification	FASP	2,104,326 parameters	20, 861,480 parameters	Acc = 49.75 %, F1 = 48.8 %
[71]	MobileNetV3Large	Detection and classification	FASP	990,214 parameters	2, 996, 352 parameters	Acc = 79.89 %, F1 = 79.7 %
[71]	DenseNet121	Detection and classification	FASP	1, 55, 750 parameters	7, 37, 504 parameters	Acc = 63.97 %, F1 = 63 %
[71]	InceptionV3	Detection and classification	FASP	2, 689, 414 parameters	21, 802, 784 parameters	Acc = 46.12 %, F1 = 45.9 %
[71]	EfficientV2S	Detection and classification	FASP	1, 317, 894 parameters	20, 331, 360 parameters	Acc = 79.19 %, F1 = 79 %
[71]	ResNet50	Detection and classification	FASP	2, 104, 326 parameters	23, 587, 712 parameters	Acc = 77.93 %, F1 = 77.9 %
[49]	Fully convolutional neural networks (FCNN)	Localization and detection	FBSP	802	201	Acc = 71 % cardiac views Acc = 81 % non-cardiac views Prec = 69 %, Recall: 80 %
[72]	FHUSP-NET	Detection and Recognition	3V, 3VT, RVOT, LVOT, 4CH	2690	670	Prec: 98.6 %, 96.1 %, 98.9 %, 100 %, 100 % for 3V, 3VT, RVOT, LVOT, 4CH
[73]	SonoNet	Localization and detection	Multiple	140,827	109,165	Prec = 77 %
[74]	multi-task learning	Detection and classification	FASP, FBSP, 4CH	2460	820	Detection: mAP = 96.38 %, 94.41 %, 95.10 % for FASP, FBSP, 4CH; Classification: Acc = 96.31 %, 94.31 %, 93.32 % for FASP, FBSP, 4CH
[75]	VGG16	Localization and classification	lateral sulcus, thalamus, choroid plexus, cavum septi pellucidi, third ventricle	4800	1153	Prec = 96.6 %,96.7 %,77.1 %,94.6 %, 72.8 % for lateral sulcus, thalamus, choroid plexus, cavum septi pellucidi, third ventricle
[75]	Faster R-CNN					Prec = 84.7 %,57 %,60.8 %,29.3 % for lateral sulcus, thalamus, choroid plexus, cavum septi pellucidi
[75]	YOLOv2					Prec = 99.6 %,97.2 %,79.9 %,94.1 %, 85.2 % for lateral sulcus, thalamus, choroid plexus, cavum septi pellucidi, third ventricle

(*continued on next page*)

Table 3 (*continued*)

Ref	Model	Task	Plane	Training set (size)	Testing Set (size)	Performance metrics
[76]	Self-supervised learning	Classification, localization, and segmentation	Multiple	140,827	109165	Prec = 89 %

TV, trans-ventricular; TC, trans-cerebellar; TT, trans-thalamic; Prec: precision; Acc: accuracy.

Table 4
AI models for fetal brain and multi-organ analysis.

Ref	Organ	Model	Training size (images)	Testing size (images)	GA (weeks)	Performance
[60]	Brain	Middle cerebral artery network (MCANet)	~	4005	28–40	DSC = 77 %, IoU = 0.63
[77]	Brain	3D convolutional regression network (3D CRN)	326	121	18–36	Acc = 93.1 %
[78]	Brain	View-based projection networks (VP-Nets)	200	45	20–29	IoU = 0.63
[79]	Brain	Deep attention network (DAN)	~	448	18–37	DSC = 77 %, Recall = 74 %
[80]	Brain	Multiple-channel and atrous convolution network (MA-Net)	718	70	12–35	DSC = 97 %, Recall = 98 %
[81]	Brain	Convolutional Neural Network (CNN)	271	36	18–22	DSC = 82 %
[82]	Brain	CNN	29419	4739	26 (normal)/29 (abnormal)	Acc = 96 %, DSC = 94 %, Recall = 94 %
[83]	Head	CNN	10	8	Second trimester	DSC = 81 %
[84]	Brain	CNN	689	122	19–30	4.1, 5.1, 4.9 for sylvian, parieto-occipital, calcarine fissures
[85]	Brain	CNN	588	146	18–20	DSC = 87 %
[86]	Brain	CNN	6201	6199	~	78.5, 79.4 for brain, common maternal fetal planes
[87]	Brain	3D CNN	885	300	14–31	DSC = 94 %
[88]	Brain	Multi-task CNN	480	48	20–25	DSC = 83 %
[89]	Brain	U-Net	138	15	20–26	DSC = 92 %
[90]	Brain	3D Encoder decoder	50	50	20–31	Acc = 99 %, IoU = 0.82
[91]	Brain	BNN	8369	869	≥14	MAE = 16.05
[92]	Head	Cycle-consistent generative adversarial network (Cycle-consistent GAN)	799	200	12–35	DSC = 94 %
[93]	Brain	Brain extraction and alignment network (BEAN)	829	356	14–30	DSC = 94 %, 93 % for extraction masks, alignment target masks
[94]	Abdomen, head	Fully convolutional network (FCN)	1588	741	19–40	DSC = 97 %, IoU = 0.96
[95]	Body, head, leg	CNN	10000	2000	16–34	Recall = 92 %
[96]	Abdomen, head, heart, spine	Curriculum learning (CL)	12808	9979	~	DSC = 96 %, Recall = 96 %
[97]	Abdomen, head	M-DINSeg	4200 (Device A)	1516 (Device B), 816 (Device C)	~	DSC = 93.58 %, 96.97 % for abdomen and head, Recall = 92.42 %, 97.43 % for abdomen and head, IoU = 0.89, 0.94 for abdomen and head
[98]	Brain, corneal face, abdomen, spine, femur	DL	496	~	19–21	Acc = 97.2 %, 86.1 %, 93.1 %, 95 %, 92.3 % for brain, corneal face, abdomen, spine, femur

Acc: Accuracy; DSC: Dice similarity coefficient; IoU: Intersection over union; MAE: Mean absolute error; GA: Gestational age.

SPRNet's impressive 99 % accuracy with 17,840 training images [53].

For classification tasks, the ensemble CNN (ECNN) introduced by Thunakala et al. [60] demonstrated outstanding performance, achieving high performance (accuracy: 95.69 %, precision: 94.02 %) for fetal US six-plane classification. Similarly, Burgos-Artizzu et al. [29] compared state-of-the-art deep convolutional neural networks (DCNN) for automatic classification of six standard planes, with DenseNet-169 achieving the highest accuracy (93.6 %). For detection tasks, the Temporal SonoEyeNet introduced by Cai et al. [63] demonstrated exceptional performance, achieving F1-scores ranging from 83.7 % to 89.9 % for detecting multiple planes in real-time video data. Building on this achievement, the same research group developed the multi-task SonoEyeNet architecture, which highlighted impressive results for FASP detection, attaining 96 % precision and recall [64]. Research groups have also explored multi-task solutions to address diverse challenges [49,70–76]. Most notably, Li et al. [72] introduced FHUSP-NET for plane-specific detection and recognition, achieving flawless precision (100 %) in identifying specific cardiac views. Zhang et al. [74] developed a comprehensive framework, achieving a detection mAP of 96.38 % and a classification accuracy of 96.31 %.

Dataset characteristics have a profound impact on model performance. Studies leveraging large datasets (>10,000 images) [46,47,49,52–54,59,61,62,66,69,71,73,76] consistently achieved higher mean accuracies compared to those utilizing medium-sized datasets (1000–10,000 images) [29,46,50,51,55–58,60,64,65,67,70,74,75] or smaller datasets (<1000 images) [48,49,68]. For instance, Qu et al. [49] investigated DL approaches such as DCNN and transfer learning alongside ML models like Cluster, SVM, and RCM for recognizing fetal standard planes across various dataset sizes. Their results showed that DL approaches (average accuracy: 88.5 %) consistently outperformed ML methods (average accuracy: 70.4 %). Notably, dataset size heavily influenced performance; DCNN's accuracy dropped by 7.7 % (from 91 % to 83.3 %) when trained on smaller datasets. These findings underscore the importance of tailoring algorithm selection to both data availability

Table 5
AI-based models for fetal brain segmentation.

Ref	Datasets	Model	DSC (%) ± std	HD (mm) ± std
[28]	HC18	Random forest	97.10 ± 2.73	1.83 ± 1.60
[99]	ISBI 2012	Difference of Gaussians	97.80 ± 1.04	2.16 ± 1.44
[99]	ISBI 2012	Random forest	97.23 ± 0.77	2.59 ± 1.14
[99]	ISBI 2012	Semi-supervised patch-based	94.45 ± 1.57	4.60 ± 1.64
[99]	ISBI 2012	Multilevel thresholding	92.53 ± 10.22	6.87 ± 9.82
[100]	HC18	Gradient Vector Flow Network (GVF-Net)	95.53 ± 3.98	2.18 ± 2.40
[101]	HC18	Mask R-CNN	97.73 ± 1.32	1.39 ± 0.82
[102]	HC18	Mini Link-Net	96.84 ± 2.89	1.72 ± 1.39
[103]	HC18	Multiscale view and DNN	92.46	3.40
[104]	HC18	Spatial Attention Fusion Network (SAF-Net)	98.05 ± 4.02	1.27 ± 0.77
[105]	HC18	Deeply Supervised Attention-Gated V-Net (DAG V-Net)	97.93 ± 1.25	1.27 ± 0.80
[106]	HC18	Regression	97.75 ± 1.32	1.32 ± 0.73
[107]	HC18	Convolutional segmentation machine (CSM)	97.61 ± 1.72	1.43 ± 0.89

DSC: Dice similarity coefficient; HD: Harsdorf distance.

and clinical priorities, striking an optimal balance between precision and recall. These results highlight the critical role of data volume in driving advancements in model performance.

4.2. Anatomical structure analysis

Table 4 provides a comprehensive comparison of recent advancements in AI models for brain and multi-organ analysis [24,60,77–98]. These comparative studies underscore the rapid progress in the field while also identifying critical challenges that must be addressed to facilitate clinical translation. The growing body of evidence highlights several promising research directions, including enhanced model generalization, improved data harmonization, and integration into clinical workflows. These advancements have the potential to significantly improve the reliability and clinical applicability of AI-assisted fetal ultrasound diagnostics in routine practice.

4.2.1. Fetal brain analysis

The evaluation of fetal brain development is crucial in prenatal care, providing insights into growth patterns and early detection of neurological abnormalities. This process, conducted through US imaging, is challenged by anatomical variations across fetuses, complicating consistent measurements. Automated analysis systems address these challenges through a four-step process: localizing structures, performing segmentation, classifying morphological features, and generating quantitative assessments. Segmentation is critical for analyzing specific regions like the middle cerebral artery, cavum septum pellucidum, cerebellum, brainstem, ventricular system, thalamus, cortical plates, and sulcal patterns. Each structure presents unique challenges due to variations in size, shape, and image characteristics, highlighting the complexity of this task.

Recent AI advancements have led to specialized segmentation algorithms for fetal brain analysis [4,20], offering tailored approaches that focus on individual structures for precise assessments. Table 5 compares AI models developed for fetal brain segmentation [28,99–107], highlighting the SAF-Net architecture by Liu et al. [104]. This model achieved impressive results in HC segmentation, with a DSC of 98.05 % and an HD of 1.27 mm on the HC18 dataset. These developments are enhancing the accuracy, reliability, and clinical relevance of fetal brain assessments.

Table 6
DL-based models for fetal HC estimation.

Ref	Model	Training set size (images)	Testing set size (images)	DSC (%)	MAE (mm)
[103]	LinkNet	878	199	91.6	~
[103]	Mini-LinkNet	878	199	92.65	~
[103]	Multi-scale mini-LinkNet	878	199	92.46	~
[104]	feature attention supervised network	999	335	97.94	~
[105]	Deeply supervised attention-gated V-Net (DAG V-Net)	999	335	97.93	1.77
[106]	Tiny-YOLO	999	335	97.75	1.90
[120]	CNN	999	335	76	~
[121]	CNN	~	~	95.49	2.44
[121]	End-to-end multi-task learning network (End-to-End MTLN)	999	297	96.84	~
[122]	Probabilistic U-Net	2848	540	98	1.81
[123]	Multi-feature pyramid U-Net (MFP-Unet)	999	335	95	~
[124]	VNet	999	297	~	±2.37
[124]	VNet-c	999	297	97.92	1.89
[125]	CNN with attention refinement and graph convolutional networks (CNN with AR and GCN)	905	94	97	~
[126]	Regression Resnet50 (Reg-ResNet50)	600	199	~	4.78
[127]	Regression VGG16 (Reg-VGG16)	600	199	~	5.46
[128]	Encoder-decoder network	800	199	92.96	2.16
[129]	Mask-R2 CNN	999	335	98	1.95
[130]	Gated Context Attention Network (GCA Net)	999	335	98.21	~
[131]	U-Net-Resnet50	600	200	98.8	1.08
[131]	LinkNet-Resnet50	600	200	~	1.15
[131]	EfficientNet architecture with mean absolute error loss function (Reg-EfficientNet-MAE loss)	600	200	~	1.83
[131]	Reg-EfficientNet-MSE loss	600	200	~	2.35
[132]	Attention U-Net (Att U-Net)	~	~	97.5	~
[132]	BiSeNet V2	~	~	97.3	~
[132]	GINet	~	~	97.2	~
[132]	global reasoning unit (GloRe)	~	~	97	~
[132]	SegFormer	~	~	97.8	~
[132]	Segmenter	~	~	97.2	~
[132]	short-term dense concatenate network (STDC)	~	~	96	~
[132]	U-Net	~	~	97.2	~

DSC: Dice similarity coefficient; MAE: Mean Absolute Error.

4.2.1.1. Gestational age (GA) estimation. While the US remains the gold standard for GA estimation, its accuracy declines significantly in later pregnancy stages, with errors exceeding 14 days due to variations in fetal growth patterns [108]. Recent AI advancements have introduced solutions through fetal brain imaging analysis [4]. Namburete et al. [109] developed a regression forest model analyzing 448 3D cranial US images to identify neurodevelopmental biomarkers, achieving an RMSE of ±6.10 days in the second and third trimesters. Burgos-Artizzu et al. [110] introduced the quantusGA model, employing supervised learning on 2D TT plane images from 1394 fetuses, outperforming traditional

biometric methods by assessing morphological brain changes. Lee et al. [111] implemented a multi-view CNN offering consistent accuracy across various populations without relying on biometric data. These advancements highlight AI's potential to improve GA estimation, particularly for populations with limited prenatal care, with models like CSM achieving 97.61 % ± 1.72 accuracy and HD of 1.43 ± 0.89 mm [107].

Recently, Patel et al. [112] investigated the application of DL technique to estimate GA from blind sweep US images. Their study introduced a novel approach that leverages free-hand sweep protocols, including the obstetric sweep protocol and blind sweeps, to improve the accessibility of ultrasound imaging in regions with limited resources and a shortage of trained personnel. The proposed methodology employs DL models to analyze US data acquired through these protocols, with a notable innovation being the integration of outlier elimination mechanisms. This enhancement aims to improve the accuracy and robustness of GA estimation by mitigating errors arising from non-standard imaging planes and data acquisition inconsistencies. Experimental results demonstrated that the approach holds significant promise for improving prenatal care in underserved areas by providing reliable GA assessments where conventional US methods are not readily available.

4.2.1.2. Fetal brain lesion diagnosis. Recent DL advancements have yielded promising outcomes in automating the diagnosis of fetal brain lesions. For example, Xie et al. [82] introduced a multi-task learning model utilizing three specialized neural networks: a U-Net with dilated convolutions for precise cranio-cerebral segmentation (achieving a DSC of 94 %), a VGG-19 network for classifying normal and abnormal brain images (96 % accuracy), and Gradient-weighted Class Activation Mapping (Grad-CAM) for lesion localization. Clinical validations confirmed that radiologists verified AI-generated lesion localizations in 61.6 % of cases. These approaches highlight DL's potential to enhance diagnostic precision, offer interpretable visualizations, and support early detection of fetal brain pathologies.

Shamya et al. [113] presented a comprehensive survey of state-of-the-art DL methodologies for the detection of fetal brain abnormalities using US data. The study critically assessed various aspects of the detection pipeline, including data acquisition protocols, image segmentation strategies, and feature extraction techniques. Among the techniques reviewed, the combination of Stacked Feature-Based Architecture (Stack FBA) and CNN demonstrated outstanding diagnostic performance, achieving an accuracy of 97.27 %, which highlights its potential applicability in routine prenatal screening and diagnostic workflows.

4.2.1.3. Multi-task approaches. Recent research highlights the effectiveness of multi-task DL frameworks for fetal brain analysis. Namburete et al. [114] developed a fully convolutional network (FCN) capable of brain localization, structural segmentation, and volume alignment in 3D fetal brain imaging, achieving an IoU of 0.82 for segmentation and an HD of 9.3 mm. Similarly, Moser et al. [87] implemented an optimized 3D U-Net architecture for brain localization and extraction, achieving a DSC of 94 % and an HD of 9.05 mm. These frameworks streamline multiple tasks into unified workflows, enhancing clinical efficiency and diagnostic consistency. By integrating localization, segmentation, and alignment, they show great promise for more standardized and efficient fetal brain assessments in routine prenatal care.

4.2.2. Multi-organ analysis

Significant progress has also been made in developing specialized algorithms for multi-organ analysis [24]. In fetal US image classification, Sridar et al. [115] designed a CNN that combines features from cropped regions and full US images, achieving an accuracy of 97 %, precision of 76 %, and recall of 75 % for 14 fetal structures. Ishikawa et al. [95] adopted VGG16 to categorize four regions with 92 % recall, while Sharma et al. [116] employed LSTM networks for spatio-temporal analysis, achieving an accuracy of 77 % across 11 categories. Innovations in automated annotation include Alsharid et al.'s [96] curriculum learning framework using NLP for image captioning (precision, recall, DSC: 96 %) and Chen et al.'s [117] unsupervised cross-domain adaptation network for unlabeled videos, achieving classification accuracies of 73 % and 89 % for 3- and 4-class.

DL has also advanced multi-organ segmentation in fetal US imaging. Gao et al. [118] developed a weakly supervised method integrating CNN and attention-gated LSTM, achieving a precision of 91 % for skull, abdomen, and heart localization. Wu et al. [119] introduced cascaded CNNs for head and abdomen segmentation, achieving a DSC of 97 %. Yang et al. [120] presented a hybrid 3D CNN-RNN framework for segmenting multiple fetal structures, including the fetus, gestational sac, and placenta, achieving a DSC of 80 % and HD of 14.13 mm. These advances highlight the potential of DL methods to improve accuracy, reduce analysis time, and enhance diagnostic consistency in prenatal care.

4.3. Fetal biometry assessment

Fetal brain biometry is integral to prenatal care, providing quantitative measurements of key brain structures through 2D and 3D imaging. These measurements are essential for determining GA, estimating fetal weight, monitoring growth, and detecting developmental abnormalities. Traditionally performed manually via US or MRI, biometry is operator-dependent, prone to variability, and time-consuming, which can impact consistency and clinical decisions. Recent advancements in AI, particularly DL, have revolutionized this field by automating biometry measurements. Parameters such as head circumference (HC), abdominal circumference (AC), and femur length (FL) are now assessed with greater precision using DL-driven approaches. These methods typically involve a two-step process: anatomical segmentation (e.g., fetal skull, femur, or abdomen) and measurement extraction using geometric algorithms. AI automation reduces human error, enhances reproducibility, and streamlines workflows, supporting more consistent and efficient prenatal care evaluations.

4.3.1. Head circumference (HC) measurement

Significant research has advanced DL algorithms to improve HC measurement accuracy [28]. DL architectures for fetal HC estimation are highlighted in Table 6 [103–106,120–132]. Skeika et al. [133] proposed V-Net with high performance (DSC: 97 %, MAE: 1.89 mm) for HC measurement. Recent innovations like GCA Net [130] and U-Net-ResNet50 [131] continue to push boundaries with DSC values above 98 %.

Direct regression methods, such as Zhang et al.'s regression-based CNN, reduce segmentation demands, estimating HC with an MAE of 4.52 mm [127]. Object detection frameworks like tiny YOLO [106] and Mask R-CNN [129] deliver high accuracy, with DSCs between 97 % and 98 % and MAEs below 2 mm. These advancements demonstrate DL's potential to match or exceed manual measurement reliability while optimizing efficiency. The shift to hybrid and regression-based techniques reflects progress in clinical deployment, balancing precision with computational demands. Current evidence suggests readiness for integrating these systems into routine prenatal care, with future developments likely focusing on real-time applications and multimodal approaches to enhance obstetric diagnostics.

4.3.2. Abdominal circumference (AC) measurement

Kim et al. [134] proposed a method that integrates special CNN and U-Net architecture, incorporating clinical decision-making processes, anatomical structures, and US image characteristics. The workflow is structured into three key stages: initial AC estimation, AC measurement, and standard plane acceptance verification. The method begins with a CNN-based classification of umbilical artery images, followed by the application of a Hough transform to obtain an initial AC estimation.

Additional CNNs are employed to estimate the position of the fetal spine and bone regions, which are then used to refine the final AC measurement. For plane validation, a U-Net segmentation network and a classification CNN jointly assess the suitability of the image for AC measurement. The proposed approach achieved a DSC of 92.55 % ± 0.83 for AC segmentation and an accuracy of 87.10 % in verifying the acceptance of the fetal abdominal standard plane. These results underscore the model's effectiveness in automating clinically relevant fetal biometry tasks.

4.3.3. Femur length (FL) measurement

Zhu et al. [135] developed a two-stage framework for femur biometry that integrates both classical ML and DL techniques. The first stage utilizes a random forest regression model that directly predicts femur endpoint coordinates without relying on image segmentation. The second stage employs a DL-based SegNet architecture, which segments the US image before applying skeletonization techniques to localize the femur endpoints. The SegNet-based method achieved a DSC of 92 % and a remarkably low MAE of 0.46 mm. Comparative analysis demonstrated that the SegNet model outperformed manual annotations in terms of consistency and robustness, whereas the random forest approach offered advantages in computational speed. These findings underscore the clinical potential of DL-based segmentation models such as SegNet for enhancing the accuracy and efficiency of fetal biometry measurements.

4.3.4. Multiple biometric parameters assessment

Recent research has developed comprehensive solutions for simultaneously assessing multiple biometric parameters. For example, Bano et al. [136] employed CNNs to jointly segment and measure HC, AC, and FL, achieving MAEs of 2.67 mm, 3.77 mm, and 2.10 mm, respectively. Płotka et al. [137] advanced this field with a multi-task architecture combining U-Net and ConvLSTM networks for fetal ultrasound videos, achieving a mean DSC of 96 % and MAEs of 2.9 mm (HC), 3.8 mm (AC), and 0.8 mm (FL). These approaches standardize measurements, reduce operator variability, and enable comprehensive evaluation of correlated growth parameters, meeting clinical accuracy requirements for fetal monitoring.

Current evidence suggests that these automated systems are transitioning from research tools to clinically viable solutions, particularly in high-volume screening environments, low-resource settings with limited specialist availability, and longitudinal growth monitoring, which requires measurement consistency. Future investigations can focus on enhancing temporal analysis, improving generalizability across populations and devices, and optimizing real-time workflows, marking a paradigm shift toward standardized, objective fetal biometry in prenatal care.

5. Clinical application of AI-driven fetal MRI

Brain segmentation is an essential component of the quantitative analysis of fetal MRI, providing critical data for assessing neurodevelopment and identifying abnormalities. This process involves three primary stages: brain extraction, brain structure segmentation, and multi-tissue annotation. Together, these stages enable a comprehensive and systematic evaluation of fetal brain anatomy and development. Recent advancements in AI, particularly DL techniques, have markedly enhanced the accuracy and efficiency of automated fetal brain segmentation from MRI scans. This section reviews state-of-the-art DL approaches for brain segmentation in fetal MRI. Studies utilizing clinical and publicly available MRI datasets [34–45] are summarized in Tables 7–1.

5.1. Segmentation using guided and unguided brain extraction methods

Guided brain extraction methods utilize preliminary processing steps to enhance segmentation accuracy. These steps often include bounding box annotations or initial brain localization, providing essential spatial context that facilitates subsequent analysis. By incorporating these preparatory measures, guided approaches ensure greater precision in segmenting complex fetal brain structures. Unguided brain extraction methods, on the other hand, have emerged as a powerful alternative for fetal MRI analysis. These techniques allow for the direct segmentation of brain structures without relying on initial localization steps such as bounding box annotations or manual initialization.

Table 7
DL-based segmentation models using guided brain extraction methods.

Ref	Model	Samples	GA (weeks)	DSC (%)
[138]	DeepCut	55 subjects	20–30	94.1
[139]	Auto-2.5D-CNN	75 vol	19–39	95.97
[140]	Auto-U-Net	75 vol	19–39	93.8
[141]	BIFSeg	18 subjects	~	95.58
[142]	DS U-Net	123 vol	~	91.68
[143]	Multi-scale FCN	88 subjects	20–30	95.8
[144]	Two-step U-Nets	35 vol	~	96
[145]	Multi-stage U-Net	DBa: 178 vol, DBb: 123 vol	DBa: 21–36, DBb: ~	DBa: 95.79, DBb: 95.45
[146]	Uncertainty-guided interactive refinement (UGIR)	105 vol	2nd trimester	91.82
[147]	DU-Nets	77 subjects	23–34	98.23
[148]	Confidence-aware Cascaded Net	239 vol	23–34	95.77

GA: Gestational age; DSC: Dice similarity coefficient.

Table 7 presents the effectiveness of various DL-based brain segmentation approaches using guided brain extraction methods, with DSC scores ranging from 91.68 % to 98.23 % across different approaches [138–148]. A notable advancement in this field is the work by Salehi et al. [139], who developed an innovative auto-context algorithm combined with CNNs. Their approach evaluated two distinct models, namely Auto-2.5D-CNN and Auto-U-Net. Auto-2.5D-CNN processes 2D image patches from axial, coronal, and sagittal planes to capture comprehensive 3D spatial information. Auto-U-Net utilizes three 2D fully convolutional networks based on the U-Net architecture, integrating outputs from multiple processing pathways to generate highly accurate brain masks.

Another significant contribution comes from Wang et al. [145], who proposed the uncertainty-guided interactive refinement (UGIR) framework. This sophisticated approach combines: a CNN for initial brain extraction, uncertainty estimation to identify potential segmentation errors, and an interactive refinement mechanism that allows for rapid correction. The UGIR framework achieved a DSC of 96 %, demonstrating the potential of hybrid human-AI collaborative systems in fetal MRI. These technological advancements enable more reliable assessment of fetal brain development by producing cleaner and more accurate segmentation results. Compared to manual or semi-automated approaches, these AI-driven techniques improve consistency, reproducibility, and efficiency, making them valuable tools in clinical settings. As these methods continue to evolve, they are expected to standardize fetal MRI analysis across different institutions and imaging platforms.

Table 8 summarizes the recently developed DL-based segmentation models using unguided brain extraction techniques [149–161]. Among these developments, Khalili et al. [151] proposed an automatic method leveraging a fully convolutional network (FCN) with a U-Net architecture to segment the intracranial volume (ICV) in fetal and neonatal MRI images. Their method demonstrated exceptional accuracy, achieving Dice Coefficients ranging from 98 % to 99 % across both fetal and neonatal scans. Another significant advancement is the comprehensive attention-based convolutional neural network (CA-Net) [155], which incorporates advanced attention mechanisms by combining spatial, channel, and scale-based components. This architecture enhances feature discrimination, enabling the network to effectively focus on brain structures while suppressing interference from surrounding

Table 8
DL-based segmentation models using unguided extraction methods.

Ref	Model	Samples	GA (weeks)	DSC (%)
[149]	Multi-scale CNN	DBax: 10 subjects, DBcor: 10 subjects, DBsag: 10 subjects	22.9–34.6	DBax: 91, DBcor: 95, DBsag: 92
[150]	Auto-U-net	285 vol	22–38	HS: 96.52, ChS: 78.83
[151]	Coarse-to-fine CNN	HS: 30 subjects, PSsb: 16 subjects, PSsbas: 16 subjects	HS: 30 ± 5, PSsb: 24 ± 1, PSbas:26 ± 1	PSsb: 93.87, PSsbas: 92.94
[152]	Fully convolutional network with U-Net (FCN-U-Net)	DBa: 45 vol, DBb: 17 vol	DBa: 23–35, DBb: 29 ± 5	DBa: 98, DBb: 94
[153]	Deep U-Net	200 vol	18–37	92.80
[154]	FS-R2-U-Net (HS), FS-Att-U-Net (PSab)	HS: 200 vol, PSab: 74 vol	18–38	HS: 89.41, PSab: 86.21
[155]	Single-input multi-output U-Net (SIMOU-Net)	HS: 200 vol, PSab: 54 vol	18–39	HS: 94.2, PSab: 91.2
[156]	Comprehensive attention-based convolutional neural network (CA-Net)	36 subjects	22–29	95.88
[159]	DeepLabV3+	60 vol	17–37	79
[158]	Residual Fusion Blocks (RFBSNet)	152 vol	22–38	HS: 97.99, ChS: 84.04
[159]	CNN auto-mask	DBa: 197 subjects, DBb: 10 subjects	DBa: 24–39, DBb: 30–36	DBa: 94, DBb: 89
[160]	Fetal generative adversarial networks (FetalGAN)	64 subjects	25–39.4	97.3
[163]	Multi-scale multi-hierarchy attention convolutional neural network (MSMHA-CNN)	DBa: 20 subjects, DBb: 10 subjects	DBa: 23–34, DBb: 26–34	DBa: 94.78, DBb: 96.71

HS: healthy subjects; ChS: challenging subjects; PS: pathological subjects; ab: normal; ax: axial; cor: coronal.
sag: sagittal; sb: spina bifida; Sbas: spina bifida after surgery; DB: database; GA: Gestational age; DSC: Dice similarity coefficient.

tissues. De Asis-Cruz et al. [159] further contributed to the field with the development of a fetal Generative Adversarial Network (FetalGAN), which utilizes a 3D U-Net generator and multi-scale loss functions for brain segmentation, achieving remarkable accuracy with a DSC of 97.3 %.

The transition from guided to unguided methods marks a pivotal milestone in fetal MRI analysis, highlighting advancements that are approaching human-level performance while delivering enhanced consistency, efficiency, and automation. Unguided methods offer substantial clinical benefits, including compatibility with high-throughput screening programs, standardized operator-independent analysis for multi-center research, and improved accuracy in longitudinal fetal growth studies. By automating and standardizing fetal brain extraction, these technologies have the potential to revolutionize prenatal diagnostics, making fetal MRI analysis more reliable, accessible, and efficient. However, unguided methods also face significant challenges, such as variability in fetal brain morphology, positioning, and gestational stage, which introduces complexity to the extraction process. Additionally, the need to differentiate fetal brain tissue from maternal anatomy and address motion artifacts caused by fetal movement poses considerable technical hurdles. As these techniques continue to evolve, they are poised to bridge the gap between research innovation and routine clinical practice, setting new benchmarks for precision in fetal neuroimaging.

5.2. Brain structure segmentation

Brain structure segmentation plays a pivotal role in fetal neuroimaging by enabling precise identification of individual anatomical regions such as the ventricles, cerebellum, and cortical plate. Unlike whole-brain extraction techniques that primarily focus on isolating the brain from surrounding tissues, this approach provides a more detailed analysis of distinct structures, each exhibiting unique morphological characteristics. The segmentation process typically begins with the reconstruction of high-resolution 3D volumes from 2D MRI slices, ensuring the spatial detail required to segment these intricate fetal brain structures accurately.

Recent advancements in DL have significantly enhanced the reliability and accuracy of fetal brain structure segmentation [7–9]. As summarized in Table 9, state-of-the-art DL-based segmentation methods

Table 9
DL models for fetal brain structure segmentation.

Ref	Model	Samples	GA (weeks)	DSC (%)
[150]	Coarse-to-fine CNN	HS: 37 subjects, PSsb: 32 subjects, PSsbas: 16 subjects	HS: 27 ± 4 PSsb: 23 ± 2 PSbas:26 ± 1	HS: 93.21, PSsb: 93.87, PSsbas: 92.94
[150]	Coarse-to-fine CNN	HS: 30 subjects, PSsb: 16 subjects, PSsbas: 16 subjects	HS: 30 ± 5, PSsb: 24 ± 1, PSbas:26 ± 1	PSsb: 93.87, PSsbas: 92.94
[163]	Auto-U-Net	HS: 15 vol, PSsb: 15 vol, PSsbas:16 vol	HS: 23, PSsb: 24.4, PSsbas: 30.1	LV HS: 91, PSsb: 94, PSsbas: 91
[164]	2D U-Net	636 vol	21–40	97
[165]	Multi-class U-Net	214 vol	22–39	CB: 85.1, LH: 94.5, RH: 94.6
[166]	TopoCP	33 subjects	21–38	CP: 70
[167]	Multi-class nnU-Net-DRO	HS: 193 vol, PSsb: 124 vol, PSab: 51 vol	19–40	HS: WM 93.8, LV 87.9, CB 94.4, PSsb: WM 90.3, LV 90.9, CB 79.7, AC: WM 90.3, LV 87.5, CB 90.6
[168]	Multi-scale 3D deep CNN	249 vol	21–38	GM: 84
[169]	Deep attentive CNN	57 subjects	16–39	CP: 87
[170]	Multi-view aggregation with test-time augmentation and fully convolutional networks (MVT-FCN)	52 subjects	22.9–31.4	CPleft: 90.7, CPright: 90.6
[171]	Swin-U-Net transformer (UNETR)	172 fetuses	late-fetal period (>25), midfetal period (≤25)	mean DSC: 92 %

HS: healthy subjects; ChS: challenging subjects; PS: pathological subjects; sb: spina bifida; sbas: spina bifida after surgery; ab: abnormal; ax: axial; cor: coronal; sag: sagittal; CB: cerebellum; WM: white matter; GM: gray matter; cGM: cortical gray matter; dGM: deep gray matter; LV: lateral ventricles; BS: brain stem; CSF: cerebro spinal fluid; eCSF, external cerebro spinal fluid; vCSF, ventricular cerebro spinal fluid; CC: corpus callosum; CP: cortical plate; BGT: basal ganglia and thalami; LH: left hemisphere; RH: right hemisphere; DB: database; GA: Gestational age; DSC: Dice similarity coefficient.

achieve DSC between 70 % and 97 %, highlighting their promising clinical applicability [150,161–171]. Among these, U-Net-based architectures have gained prominence for their superior performance, particularly in the segmentation of the ventricular system across both normal and pathological cases. These models excel even in challenging scenarios such as cases of ventricular dilation—a common clinical condition that complicates traditional segmentation techniques.

To address persistent challenges in this domain, researchers have introduced innovative methodologies. A noteworthy contribution was made by de Dumast et al. [164], who introduced a topological loss function, termed TopoCP, into the U-Net architecture to ensure morphological consistency during the segmentation of complex structures like the cortical plate. This innovation produces anatomically plausible results that align closely with the biological realities of fetal brain development. In addition, distributional robust optimization techniques have been developed to address data imbalances in pathological cases. For example, the multi-class nnU-Net-DRO framework effectively reweighs underrepresented samples during training, leading to significant improvements in segmentation performance for white matter, ventricles, and cerebellum across diverse clinical datasets [165].

The field has seen significant advancements in multi-scale and multi-plane analysis techniques. Multi-scale 3D deep convolutional neural networks (multi-scale 3D deep CNNs) effectively integrate automated processing with human-in-the-loop refinement, providing a valuable solution for cortical gray matter segmentation, where precision is crucial [166]. Hong et al. [168] developed a hybrid model that combines multi-view aggregation, test-time augmentation, and fully convolutional networks (MVT-FCN) to improve cortical plate segmentation accuracy. Their approach achieved DSC scores of 90.7 % and 90.6 %, along with mean surface distance errors of 0.182 mm and 0.185 mm for the left and right cortical plates, respectively.

These technological advancements are transforming both clinical practice and research into fetal neurodevelopment. Enhanced segmentation capabilities enable the early detection of subtle neurodevelopmental anomalies, more accurate quantitative monitoring of brain maturation, and improved surgical planning for fetal interventions. As these methods continue to advance, they are setting new benchmarks for precision in prenatal brain assessment, significantly enhancing diagnostic capabilities and deepening our understanding of fetal neurodevelopment. Future innovations in this field will likely focus on integrating topological constraints, attention mechanisms, and multi-plane analysis to refine segmentation accuracy further and broaden clinical applicability.

5.2.1. *Multi-tissue annotation*

Multi-tissue annotation represents a groundbreaking advancement in fetal brain MRI, offering voxel-level classification of developing neural tissues, including white matter, gray matter, and cerebrospinal fluid. This sophisticated approach provides crucial insights into normal neurodevelopment and enables early detection of pathological changes. By surpassing conventional whole-brain segmentation techniques, multi-tissue annotation delivers unparalleled precision in analyzing distinct anatomical regions. However, its widespread clinical adoption is hindered by challenges such as the scarcity of high-quality annotated datasets, variability in imaging protocols across institutions, and the absence of standardized annotation frameworks. The Fetal Tissue Annotation and Segmentation Challenge (FeTA) has been instrumental in addressing these limitations. By establishing consistent evaluation benchmarks and releasing open-access datasets, FeTA has catalyzed progress in developing robust DL algorithms for fetal brain analysis.

Significant advancements in DL-based multi-tissue segmentation methods, as summarized in Table 10 [156,170–174], showcase notable progress in the field. Khalili et al. [170] introduced a U-Net-based architecture for automated multi-tissue annotation, achieving an average DSC of 88.4 %. Subsequent advancements have expanded segmentation capabilities to include additional tissue classes, enhancing accuracy for diagnostically critical structures such as white matter and ventricles [173]. Building on this foundation, Zhao et al. [156] applied a multi-class 3D U-Net model, achieving superior performance over traditional atlas-based approaches, particularly in complex clinical populations. More recently, hybrid architecture integrating contextual transformers with convolutional neural networks (CoT-CNN) has demonstrated remarkable segmentation accuracy while maintaining computational efficiency [174]. The clinical implications of these innovations are substantial. Enhanced multi-tissue annotation improves the sensitivity of detecting subtle neurodevelopmental anomalies and provides quantitative biomarkers for monitoring brain maturation. These capabilities hold value in high-risk pregnancies and research on fetal neurological disorders, supporting timely interventions and improved clinical outcomes.

Table 10
DL models for multi-tissue annotation.

Ref	Model	Samples	GA (weeks)	DSC (%)
[156]	Multi-class 3D U-Net	106 vol	24.4–39.4	Average: 89.7, CSF: 92.2, cGM: 82.8, WM: 90.8, dGM: 88.4, CB: 93.5, BS: 90.2
[170]	Multi-class U-Net	12 subjects	22.9–34.6	Average: 88.4, CB: 80.2, BGT: 88.9, vCSF: 87.5, WM: 92.2, BS: 93.0, cGM: 82.9, eCSF: 94.3
[171]	Multi-class U-Net	15 subjects	22.6–33.4	Average: 86
[172]	3D U-Net	HS: 168 vol, PSsb: 96 vol	20–35	Average: 74.8, WM: 91.5, LV: 90.7, CB: 89.6, eCSF: 75.3, cGM: 56.6, dGM: 71.4, BS: 61.5, CC: 62.0
[173]	Multi-class nnU-Net	294 subjects	19.6–38.9	Young (GA < 32): 89.3, Older (GA ≥32): 91.6
[174]	Contextual transformer with convolutional neural networks (CoT-CNN)	80 vol	20–35	Average: 83.79, eCSF: 85.49, GM: 71.20, WM: 89.73, LV: 86.38, CB: 85.75, dGM: 84.64, BS: 83.31

HS: healthy subjects; PS: pathological subjects; sb: spina bifida; CB: cerebellum; WM: white matter; GM: gray matter; cGM: cortical gray matter; dGM: deep gray matter; LV: lateral ventricles; BS: brain stem; CSF: cerebro spinal fluid; eCSF, external cerebro spinal fluid; vCSF, ventricular cerebro spinal fluid; CC: corpus callosum; GA: Gestational age; DSC: Dice similarity coefficient.

Future research will prioritize enhancing the generalizability of models across diverse populations and imaging protocols, alongside improving integration with clinical workflows. The development of robust quality control mechanisms and standardized reporting frameworks will be vital for successful clinical translation. Collaborative efforts, like those spearheaded by FeTA, will continue to advance innovation while ensuring clinical relevance. As the field evolves, these advancements promise to redefine precision in fetal brain MRI, elevating diagnostic capabilities and deepening our understanding of fetal neurodevelopment.

5.2.2. *Miscellaneous applications*

Table 11 summarizes several DL models developed for miscellaneous applications in fetal MRI analysis, highlighting their contributions to improving image quality, reconstruction, and quantitative assessment. Ebner et al. [150] developed a comprehensive coarse-to-fine CNN model for automated isotropic reconstruction of fetal brains from multiple 2D scans. Their sophisticated pipeline incorporates a two-stage CNN-based

Table 11
DL models for miscellaneous applications.

Ref	Model	Samples	GA (weeks)	DSC (%)
[150]	Coarse-to-fine CNN	HS: 37 subjects, PSsb: 32 subjects, PSsbas: 16 subjects	HS: 27 ± 4, PSsb: 23 ± 2, PSsbas:26 ± 1	HS: 93.21, PSsb: 93.87, PSsbas: 92.94
[150]	Coarse-to-fine CNN	HS: 30 subjects, PSsb: 16 subjects, PSsbas: 16 subjects	HS: 30 ± 5, PSsb: 24 ± 1, PSsbas:26 ± 1	PSsb: 93.87, PSsbas: 92.94
[162]	2D U-Net	636 vol	21–40	97
[163]	Multi-class U-Net	214 vol	22–39	CB: 85.1, LH: 94.5, RH: 94.6

HS: healthy subjects; PS: pathological subjects; sb: spina bifida; sbas: spina bifida after surgery; CB: cerebellum; LH: left hemisphere; RH: right hemisphere; GA: Gestational age; DSC: Dice similarity coefficient.

brain extraction followed by robust super-resolution reconstruction algorithms. Validation results showed high performance, with DSC scores ranging from 92.94 % to 93.87 % in both healthy and pathological cases, including subjects with spina bifida and post-surgical patients. This approach significantly advances the ability to obtain high-quality 3D reconstructions from clinically acquired 2D scans, addressing a long-standing challenge in fetal MRI.

Li et al. [162] proposed an integrated 2D U-Net framework that simultaneously addresses brain extraction, super-resolution reconstruction, and 4D brain mapping for developmental analysis. Their model achieved exceptional performance (DSC: 97 %) while enabling quantitative assessment of brain development across gestational ages 23–36 weeks in a Chinese cohort. This work demonstrates the potential for comprehensive fetal brain analysis within a unified DL architecture, streamlining multiple essential processes into a single automated workflow.

Avisdris et al. [163] applied a multi-class U-Net model to compute various fetal brain measurements from MRI scans. Their approach obtained strong segmentation performance for cerebral hemispheres (LH: 94.5 %, RH: 94.6 %) and satisfactory results for more challenging structures like the cerebral biparietal cortex (85.1 %). This application highlights how DL models can automate quantitative morphometric analysis in fetal neuroimaging, facilitating objective and standardized assessments.

The success of these approaches suggests promising directions for future research, particularly in developing more comprehensive pipelines that combine image reconstruction with advanced analysis capabilities. Such integrated systems could significantly enhance the clinical utility of fetal MRI while reducing processing time and improving standardization across institutions. As fetal MRI continues to evolve, DL-driven innovations are expected to transform clinical workflows, making fetal brain imaging more accessible, accurate, and efficient.

6. Clinical applications of AI-driven multimodal imaging

This section highlights recent advancements in AI-driven multimodal imaging for improving the diagnosis and management of pregnancy-related disorders and fetal health. In 2020, Van der Knoop et al. [10] evaluated the diagnostic efficacy of multiplanar neurosonography and MRI in high-risk fetuses with suspected brain abnormalities. Their findings indicated that multiplanar neurosonography identified a higher number of fetal brain anomalies compared to conventional axial US and MRI, suggesting that MRI did not provide additional diagnostic benefits in this context. The authors recommend multiplanar neurosonography as the preferred imaging modality for detecting acquired brain anomalies in at-risk fetuses due to its superior resolution and diagnostic accuracy.

Recently, Torrents-Barrena et al. [175] investigated the integration of radiomics and DL techniques for segmenting fetal and maternal anatomy using both MRI and 3D US data. The study involved extracting 94 radiomic features to characterize intrauterine tissues, followed by the application of feature selection algorithms to identify the most informative features for each anatomical structure. These selected features were then used to train a support vector machine model aimed at achieving precise segmentation. In parallel, the authors evaluated several DL-based segmentation frameworks, including DeepLabV3+, BiSeNet, PSPNet, and Tiramisu, validating their performance on a dataset of 60 axial MRI and 3D US images from both pathological and clinical cases. The findings demonstrated that combining selected radiomic features with advanced DL architectures significantly enhances segmentation accuracy for fetal and maternal tissues. This work highlights the potential of hybrid radiomics-DL models in improving image segmentation and supports the development of technologies for fetal surgical planning.

More recently, Heinsalu et al. [176] investigated the predictive power of combining structural and functional MRI parameters with US data for preterm birth risk assessment. Using a supervised machine learning model trained on data from 58 preterm and 217 term-born neonates, their best-performing algorithm achieved an R^2 of 0.67 and correctly classified 92 % of preterm cases. These results demonstrate the potential of multimodal imaging and AI to enhance preterm birth prediction, enabling timely clinical interventions for high-risk pregnancies.

Kothinti et al. [177] developed an AI-based system for detecting fetal brain and heart abnormalities using combined US and MRI data. The proposed model, which employed CNNs and transformer architectures, outperformed conventional diagnostic methods in accuracy, sensitivity, and specificity. The fusion of imaging features from both modalities improved the detection of subtle anomalies, reducing inter-observer variability and standardizing prenatal assessments. This study exemplifies how AI can refine diagnostic precision in fetal medicine.

Despite these advances, there is a notable gap in the literature and clinical applications when it comes to true multimodal fusion of fetal US and MRI using AI. Most studies focus on improving each modality independently or employ image translation models to bridge the modality gap. However, these are often proof-of-concept studies and have not yet been translated into integrated diagnostic tools used in routine fetal imaging. Few efforts have tackled challenges such as spatial and temporal registration of MRI and US, domain adaptation, and multimodal feature-level fusion. The heterogeneity in acquisition protocols, timing of scans, and patient motion further complicates this task. As a result, while the potential for multimodal AI systems to combine the speed of US with the resolution of MRI is immense, the current clinical utility of such systems remains largely aspirational.

7. Discussion

7.1. Current challenges in AI-driven fetal imaging

AI integration in fetal imaging shows great promise, yet several barriers hinder its clinical adoption. A primary limitation is the scarcity of publicly available, high-quality, and diverse datasets. Most studies utilize small to medium-sized datasets, often comprising fewer than 1000 images, which lack adequate representation of pathological conditions and diverse demographics. This results in a reduced ability for AI models to generalize across varied patient populations, leading to overfitting to specific data distributions. Moreover, many datasets focus on segmentation or single-image classification, limiting the development of robust AI algorithms capable of supporting complex clinical decisions.

Technical challenges vary by imaging modality. In fetal US, manual preprocessing steps, such as ellipse fitting for biometric measurements,

introduce human error and inefficiency. Current research is disproportionately focused on head circumference estimation, partly due to the availability of datasets like HC18. However, other critical biometric markers, such as abdominal circumference and femur length, remain underexplored. Conversely, fetal MRI faces distinct limitations. Motion artifacts caused by fetal and maternal movement, variability in acquisition protocols (e.g., differences in slice thickness across institutions), and a lack of temporal alignment complicate the development of AI models. These models must adapt to continuous anatomical variation during fetal development, yet current segmentation performance—though promising for healthy third-trimester brains—declines significantly for earlier gestational ages (<24 weeks) and in the presence of pathology.

Moreover, multimodal integration of the US and MRI presents additional hurdles, including spatial misalignment between modalities, heterogeneity in institutional imaging protocols, and the high computational burden of processing high-resolution 3D data. A lack of publicly available datasets containing paired US-MRI acquisitions further limits the development of multimodal algorithms and benchmarking opportunities. Ethical considerations are particularly salient in prenatal AI applications, as diagnostic errors could have profound clinical and psychological consequences. Therefore, models must be rigorously validated and incorporate explainability tools to ensure alignment with established clinical markers. Despite technical progress, the translation of AI-assisted multimodal fetal imaging into clinical workflows remains minimal. Clinical implementation will only be realized when models demonstrate reliability, interpretability, and adaptability across real-world scenarios.

7.2. Generalizability and barriers to real-world implementation

Generalizability remains a significant barrier to the clinical deployment of AI in fetal imaging. Variability in scanners, such as differences in US probe frequencies or MRI field strengths, acquisition protocols, and operator expertise introduces heterogeneity, leading to performance degradation when models are applied beyond their training settings. Domain shifts due to gestational age variation, fetal position, maternal body mass index (BMI), and underrepresented pathological cases further challenge the robustness of AI models. Although domain adaptation techniques and transfer learning strategies are increasingly explored, their integration into fetal imaging pipelines remains limited.

A significant but often overlooked issue is the heavy reliance on traditional performance metrics, such as accuracy and DSC, to evaluate AI models. While these metrics provide a baseline for technical validation, they fail to reflect the real-world complexities of clinical environments. High accuracy, for example, does not guarantee improved clinical outcomes or workflow integration. In practice, clinicians require AI systems that not only classify, or segment correctly but also improve diagnostic confidence, reduce inter-observer variability, and assist in decision-making under time constraints. Robustness to scanner variability, demographic diversity, and imaging artifacts, factors often excluded in benchmark evaluations, are essential for trustworthy deployment. Therefore, future work should complement traditional metrics with clinically meaningful evaluations, such as reductions in diagnostic delays, improvements in prenatal counseling, and usability studies that incorporate clinician and patient feedback.

Additionally, technical demands related to real-time inference, compatibility with picture archiving and communication systems, and integration into existing hospital IT infrastructure remain nontrivial. For multimodal systems that fuse US and MRI, spatial and temporal misalignment further complicates deployment, especially outside controlled research environments. Many current deep learning models function as opaque "black boxes," limiting interpretability and user trust. While explainability techniques like saliency maps or class activation mappings exist, few are tailored for high-stakes prenatal settings, where interpretability must align with clinical reasoning. The lack of prospective, multicenter trials and standardized benchmarks continues to stall regulatory approval and clinical confidence in these technologies.

Addressing these challenges will require a concerted effort to validate AI systems in diverse, real-world settings, prioritize inclusive datasets, and redefine success metrics to align with actual clinical impact. Only by bridging the gap between technical performance and clinical utility can AI in fetal imaging move beyond the proof-of-concept stage toward meaningful, safe, and equitable implementation in prenatal care.

7.3. Future research directions

To address these challenges, future research must focus on several key areas. First, there is a critical need for larger and more diverse datasets. Collaborative initiatives and open-access datasets, such as the PSFHS and ACOUSLIC-AI, are essential to create a diverse repository for fetal imaging. The PSFHS dataset provides extensive US data with segmentation capabilities, while the ACOUSLIC-AI dataset focuses on abdomen segmentation in blind sweeps. These datasets not only broaden research opportunities but also serve as benchmarks for evaluating AI models. To maximize their utility, standardized data acquisition protocols and annotation guidelines are needed to ensure consistency across institutions and imaging modalities. The inclusion of multi-modal datasets, incorporating both US and MRI data, will facilitate the development of more robust AI models. Additionally, standardized data acquisition protocols and clear annotation guidelines are needed to ensure consistency across institutions and imaging modalities.

Methodological innovation remains critical. In the fetal US, end-to-end DL pipelines could reduce reliance on manual preprocessing, minimizing errors and improving efficiency. For fetal MRI, developing gestational age-adaptive architectures, such as transformer-based networks, will enhance segmentation accuracy across developmental stages. In multimodal AI, attention-based fusion mechanisms offer promise in integrating US and MRI data for improved anomaly detection. Synthetic data generation, particularly via generative adversarial networks (GANs), may help address class imbalance and data scarcity, especially for rare conditions.

Clinical translation requires rigorous validation of AI systems through prospective, randomized trials comparing AI-assisted and standard workflows. Real-time applications, such as scan-plane guidance and motion correction, have the potential to streamline procedures and reduce inter-operator variability. Privacy-preserving federated learning frameworks may facilitate multicenter collaboration without compromising patient confidentiality, promoting model robustness and scalability.

Ethical considerations are crucial in the deployment of AI in fetal imaging, especially given the sensitive nature of prenatal care. Ensuring patient privacy and data security is paramount, with techniques like federated learning facilitating collaborative research while maintaining confidentiality. Furthermore, to avoid perpetuating health disparities, bias in AI models must be addressed by ensuring that training datasets reflect diverse demographic groups, including variations in race, ethnicity, maternal health factors such as BMI and age, and socioeconomic status. Transparency in AI decision-making is equally important, with explainable AI methods being necessary to allow clinicians to trust and understand AI-generated results. Establishing clear regulatory frameworks and regulatory sandboxes can ensure that AI technologies are tested and refined in real-world clinical environments, supporting their safe, effective, and equitable deployment. Addressing these ethical concerns is essential for the responsible integration of AI into clinical practice, ensuring that it benefits all patient populations fairly and effectively.

Ultimately, AI-driven multimodal fetal imaging could revolutionize prenatal care by enabling earlier and more accurate diagnoses, improving surgical planning for in utero interventions, and facilitating

longitudinal monitoring of fetal development. To achieve this, future work must focus on building harmonized multimodal datasets, developing robust cross-modality registration algorithms, and conducting prospective validation in diverse populations. AI models must be designed with inclusive and generalizability to mitigate bias and promote equitable care.

The future of AI in fetal imaging depends on a coordinated effort among researchers, clinicians, and regulatory bodies. Establishing international consortia to curate multimodal datasets will be pivotal for fostering innovation and inclusivity. Hybrid models that combine DL with clinical expertise can enhance interpretability and trustworthiness. Moreover, designing AI systems that augment, rather than replace, clinician expertise, through context-aware prompts and explicit uncertainty quantification, will ensure their seamless integration into clinical workflows. By addressing these challenges and advancing research in the proposed directions, AI can transition from a promising research tool to a transformative technology in fetal medicine. This evolution holds the potential to reduce diagnostic variability, improve patient outcomes, and establish AI as an indispensable component of prenatal care.

Although this study compiles a substantial body of literature on AI applications in fetal imaging, we did not conduct a formal meta-analysis due to the considerable heterogeneity across studies. Variations in imaging modalities (e.g., 2D/3D US, T2-weighted MRI), gestational age ranges, AI architectures, evaluation protocols, and outcome definitions, as well as inconsistent reporting of metrics such as confidence intervals and validation strategies, posed significant barriers to meaningful statistical synthesis. To maintain analytical rigor and clinical relevance, we instead employed a structured narrative approach, supplemented by tabulated performance metrics for side-by-side comparison. We encourage future research to adopt more standardized methodologies and consistent reporting practices, which would facilitate robust meta-analytic efforts within more homogeneous subdomains of fetal imaging.

8. Conclusion

Advancements in AI are set to reshape the landscape of fetal imaging, offering innovative solutions to overcome longstanding challenges in prenatal diagnostics. Despite significant progress, obstacles such as limited access to diverse and robust datasets, inefficiencies in computational methods, and concerns surrounding ethical practices continue to hinder widespread adoption of AI technologies in clinical settings. This study examines the integration of US and MRI as complementary modalities that capitalize on their respective strengths—US for accessible and real-time imaging and MRI for detailed visualization of fetal anatomy. Addressing these challenges requires focused efforts on developing standardized datasets with broad demographic representation, refining algorithmic models to account for anatomical variability and reduce computational barriers, and creating ethical frameworks to ensure transparency and data privacy. Additionally, fostering interdisciplinary collaboration and introducing AI-powered tools, such as real-time systems for diagnostics, can significantly enhance precision and workflow efficiency in fetal imaging. By proposing a strategic pathway for research and implementation, this study emphasizes AI's potential to revolutionize prenatal care through improved diagnostic accuracy and reliability while minimizing diagnostic disparities.

CRediT authorship contribution statement

Lulu Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Mostafa Fatemi:** Writing – review & editing. **Azra Alizad:** Writing – review & editing.

Ethics statement

This study complies with ethical standards, involving no human or animal subjects. All data were sourced ethically or generated by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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